18 Show that $\dfrac{1}{12^{-\frac{3}{2}}}$ can be written in the form $a\sqrt{b}$ where a is an integer and b is a prime number.

$1 \div 12^{\frac{-3}{2}} = 12^{\frac{3}{2}} = (\sqrt{12})^3$

1 mark
Negative index: take the reciprocal.

$= 12\sqrt{12} = 12 \times 2\sqrt{3}$

$= 24\sqrt{3}$

$a = 24$; $b = 3$, which is prime.

2 marks total
Power 3/2 means square-root then cube; simplify the surd.

(Total for Question 18 is 3 marks)

DO NOT WRITE IN THIS AREA